Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 70 percent training data and the rest for testing. Each experiment was repeated 5 times with different random seeds.

2 Results

Figure 1 shows the f1 score of Transformer by dataset.

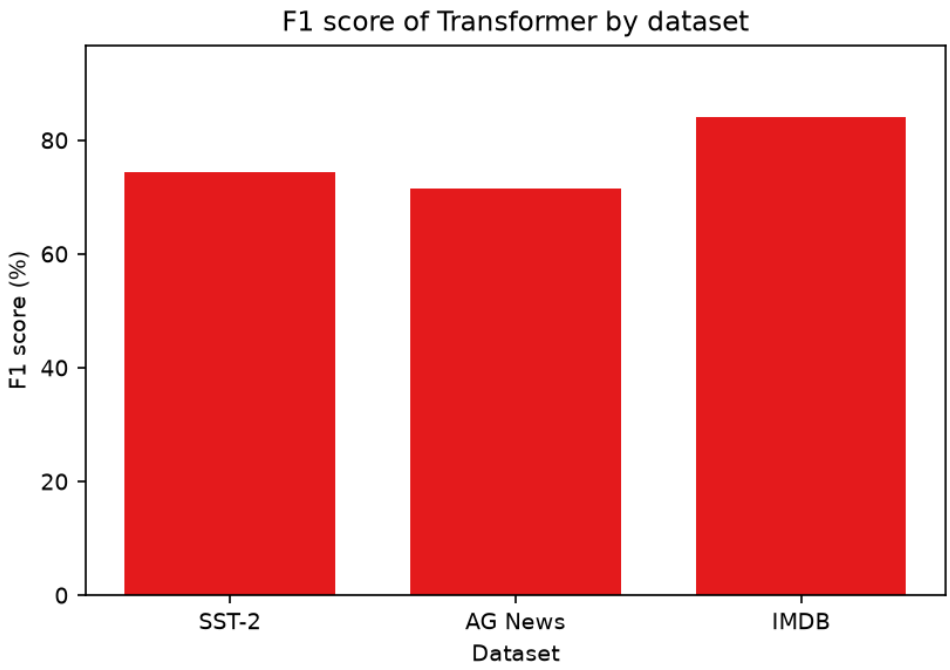


Figure 1: F1 score of Transformer by dataset for the models in this experiment.

Transformer reaches 119.9% F1 score on AG News. Transformer reaches 71.5% F1 score on AG News (Figure 1).

Figure 2 shows the accuracy of Ours by dataset.

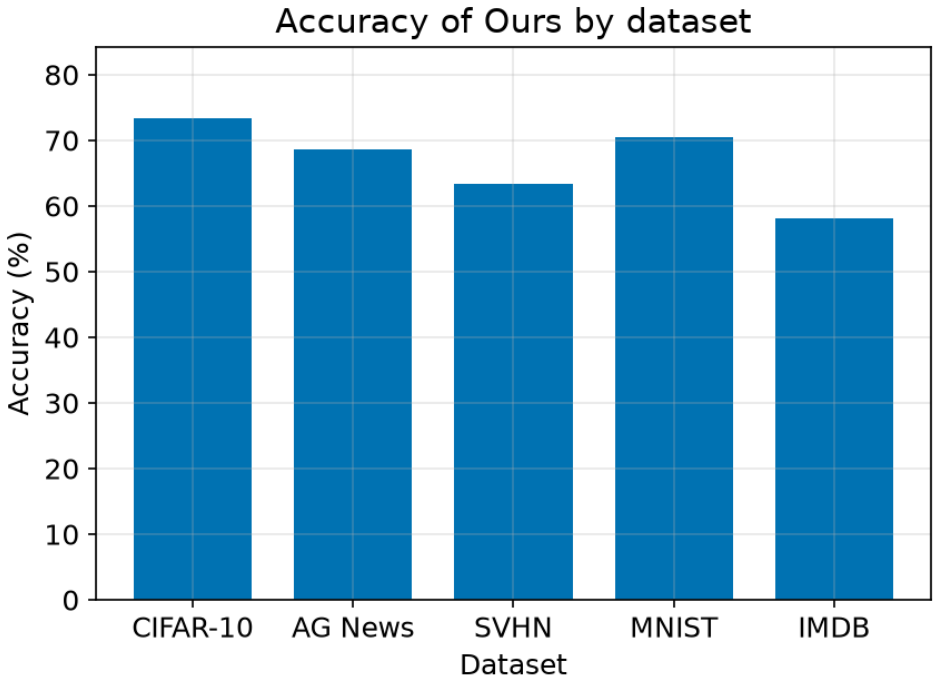


Figure 2: Accuracy of Ours by dataset for the models in this experiment.

Ours achieves 71.8% accuracy on SVHN. Ours achieves 63.4% accuracy on SVHN. Ours achieves 70.5% accuracy on MNIST (Figure 2).

3 Discussion

On AG News, Naive Bayes obtains an accuracy of 83.5%. The differences between the models are discussed in the next section.